AIRLINE FLIGHT DELAYS:
SYSTEM DYNAMICS & WEATHER

Abstract: This project analyzes U.S. airline delay data to evaluate patterns in delay frequency, severity, and causes, as well as the predictability of delays. Results indicate that weather accounts for a minority of delays, while system-level factors dominate. A focused case study of NYC airports highlights an increase in the severity of weather-related disruptions over time.

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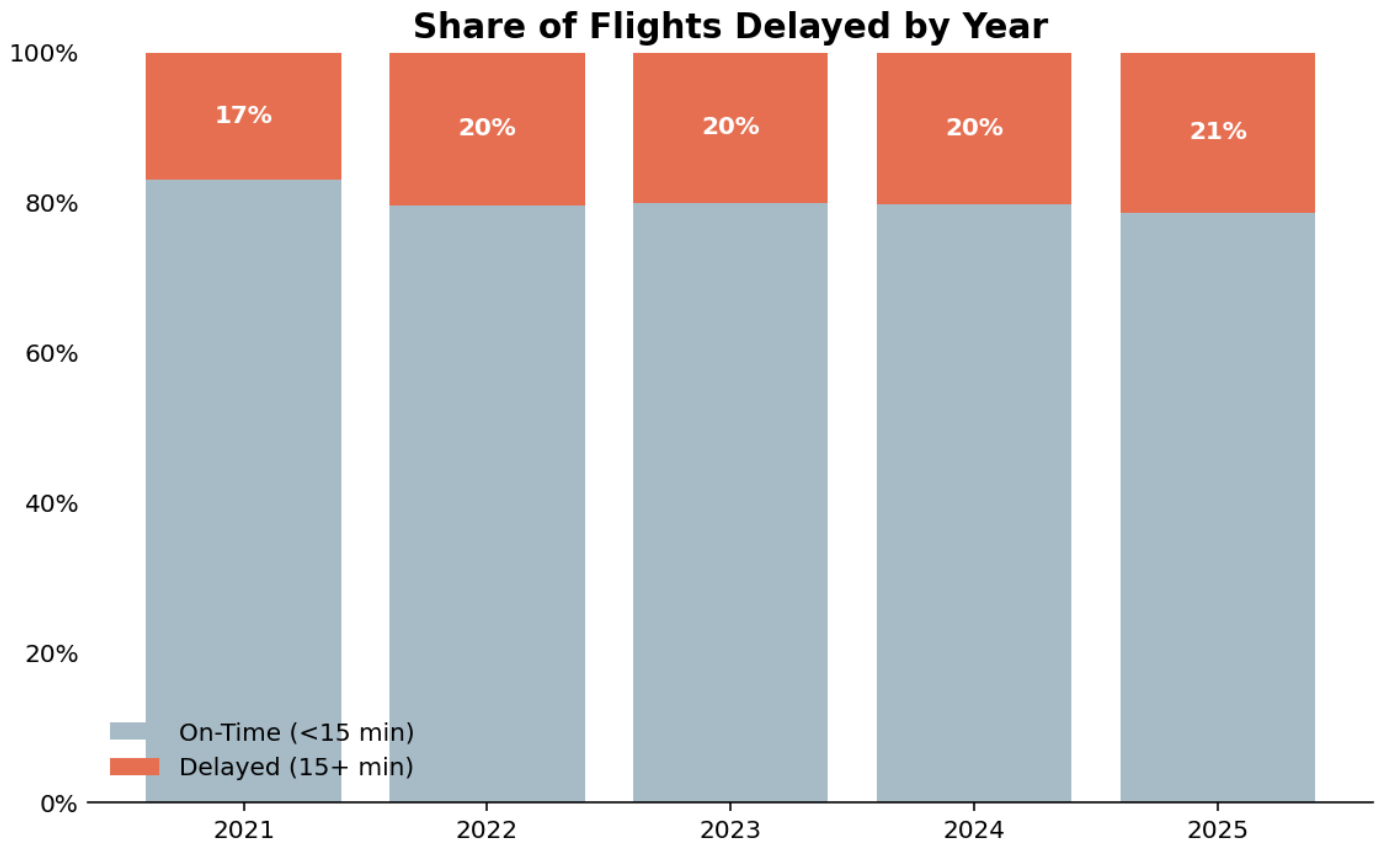
https://github.com/amkaslewicz/Airline-Flight-Delays

DELAY RATE — OVERVIEW



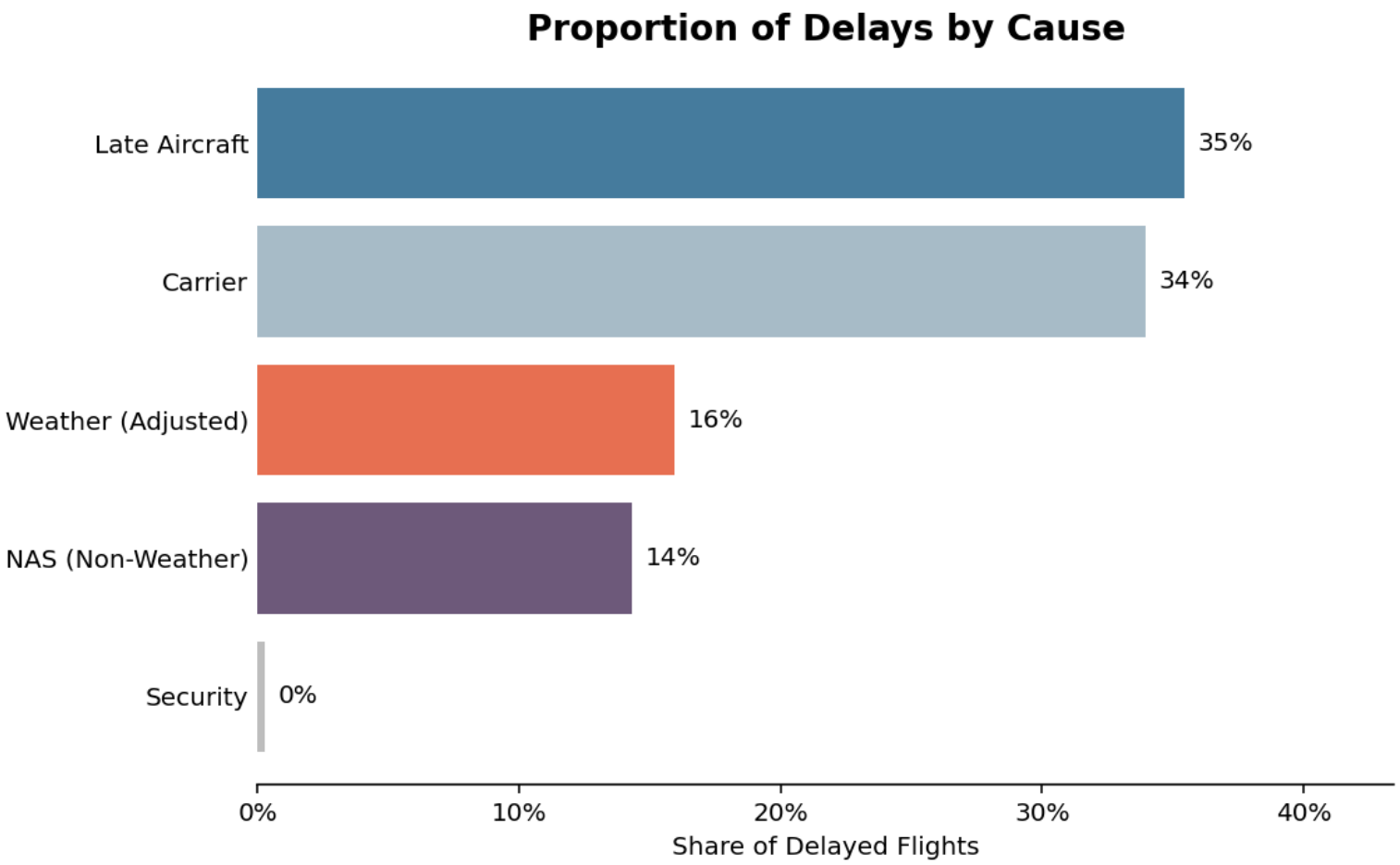
Average delay rate

19.9% of flights

Avg. delay when delayed

70 minutes

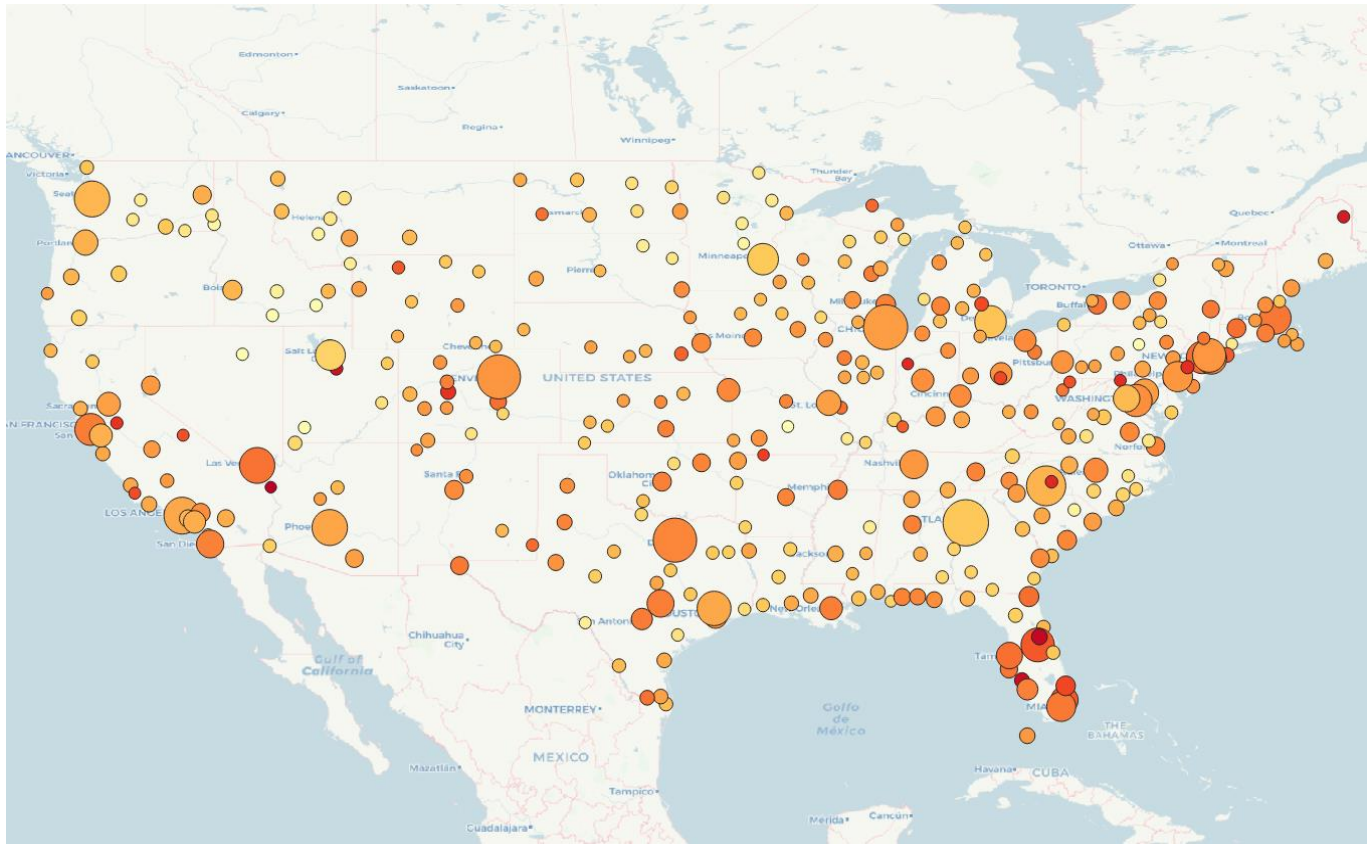
CAUSES OF DELAYS



Key Takeaway

Late-arriving aircraft and carrier issues drive the majority of delays — both are operational factors within airlines' control.

WHERE DELAYS HAPPEN



Delay Probability

- 0.12
- 0.14
- 0.16
- 0.18
- 0.2
- 0.22
- 0.24
- 0.26
- 0.28
- 0.3
- 0.32

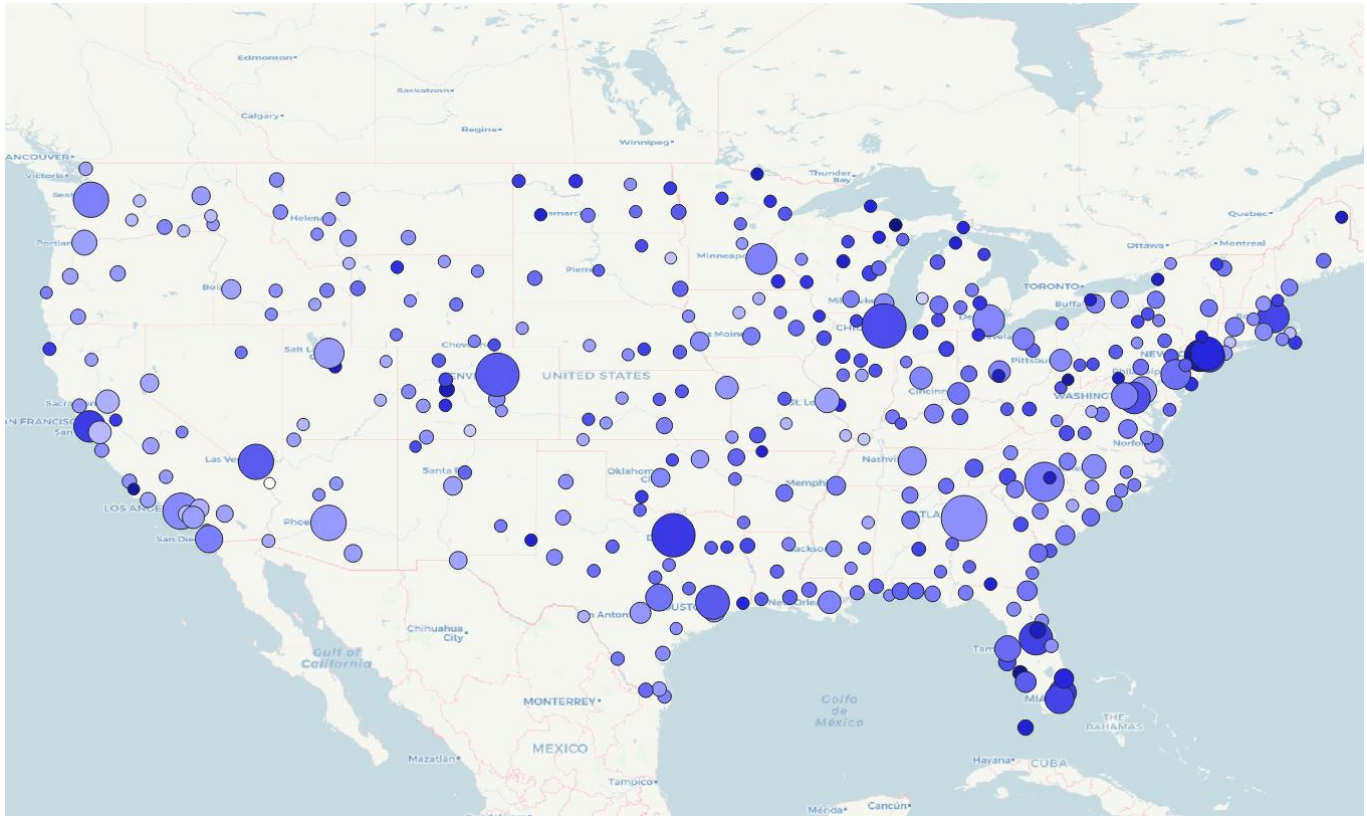
Airport Size

- 200000
- 400000
- 600000
- 800000
- 1e+06
- 1.2e+06
- 1.4e+06
- 1.6e+06
- 1.63059e+06

Key Takeaway

Delay probability is highest at large hub airports in the Northeast and Southern Florida.

WHERE WEATHER MATTERS



Weather Score
(Frequency & Severity)

- -2
- -1
- 0
- 1
- 2
- 3
- 3.5

Airport Size

- 200000
- 400000
- 600000
- 800000
- 1e+06
- 1.2e+06
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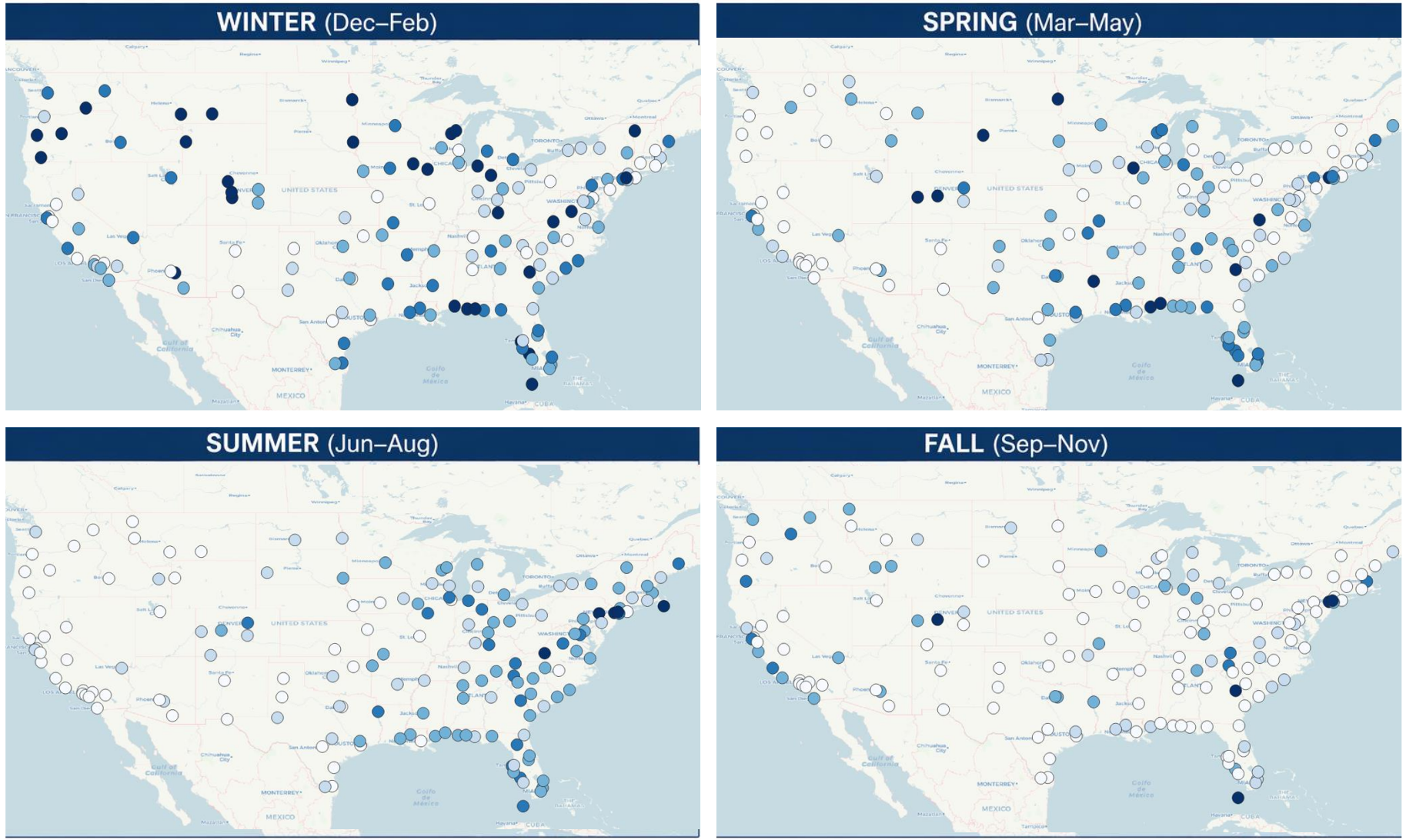
Key Takeaway

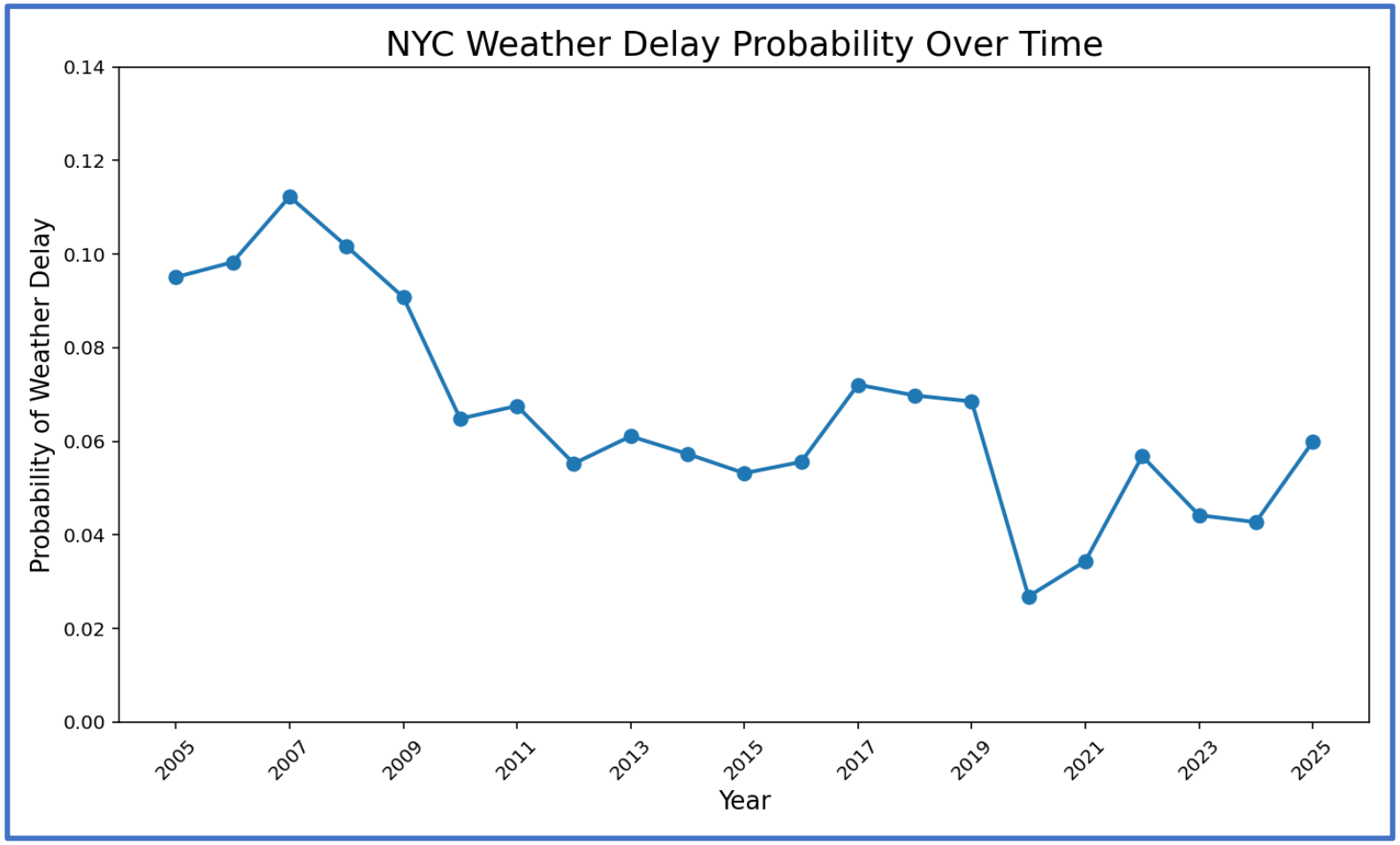
Weather-driven delays concentrate in Southern Florida, Northeast and Great Lakes hubs. Winter storms and coastal systems amplify risk at already-busy airports.

SEASONAL WEATHER DELAY TRENDS



NYC CASE STUDY – WEATHER DELAYS

NYC AIRPORTS (JFK, LGA, EWR) 2005–2025



Key Takeaway

NYC weather delay probability has trended down from ~11% in 2007 to ~5–6% today

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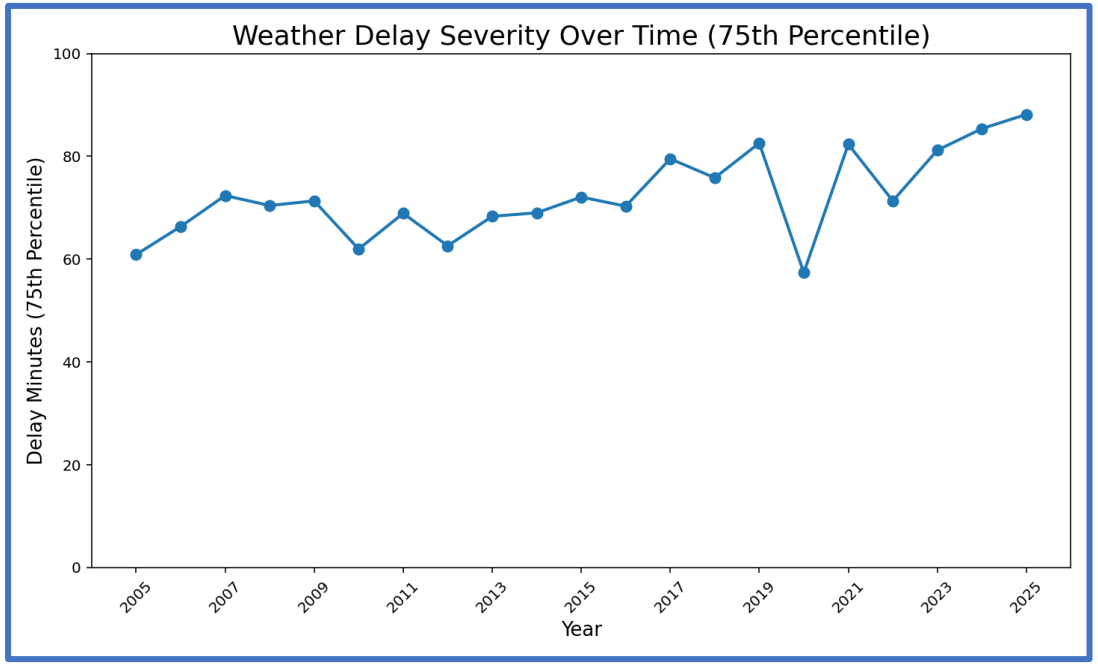
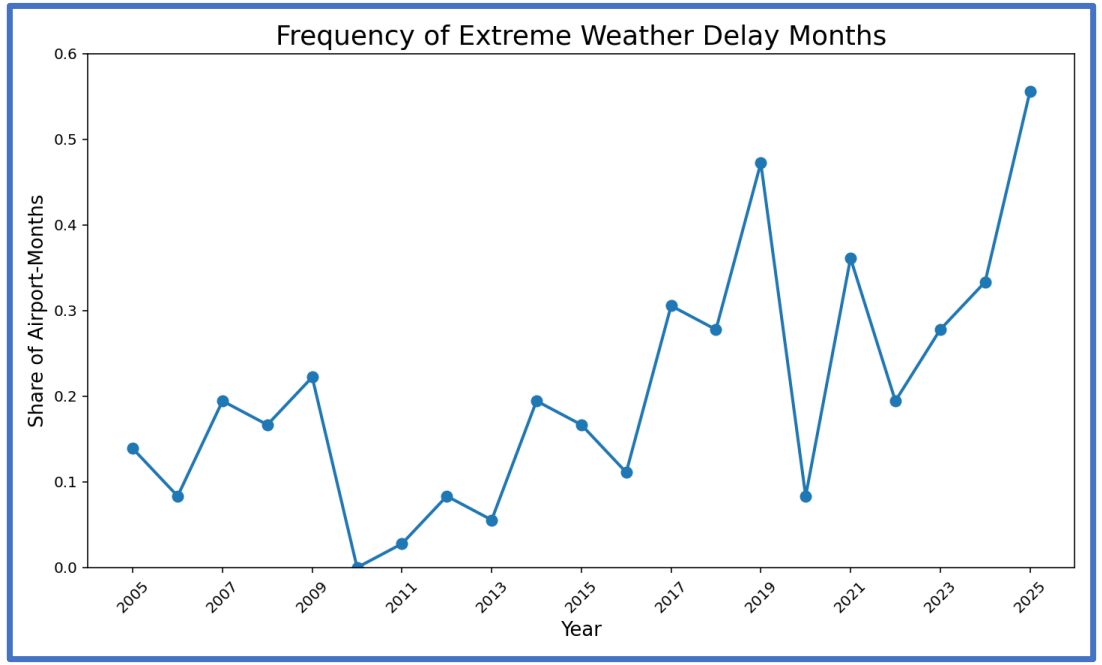
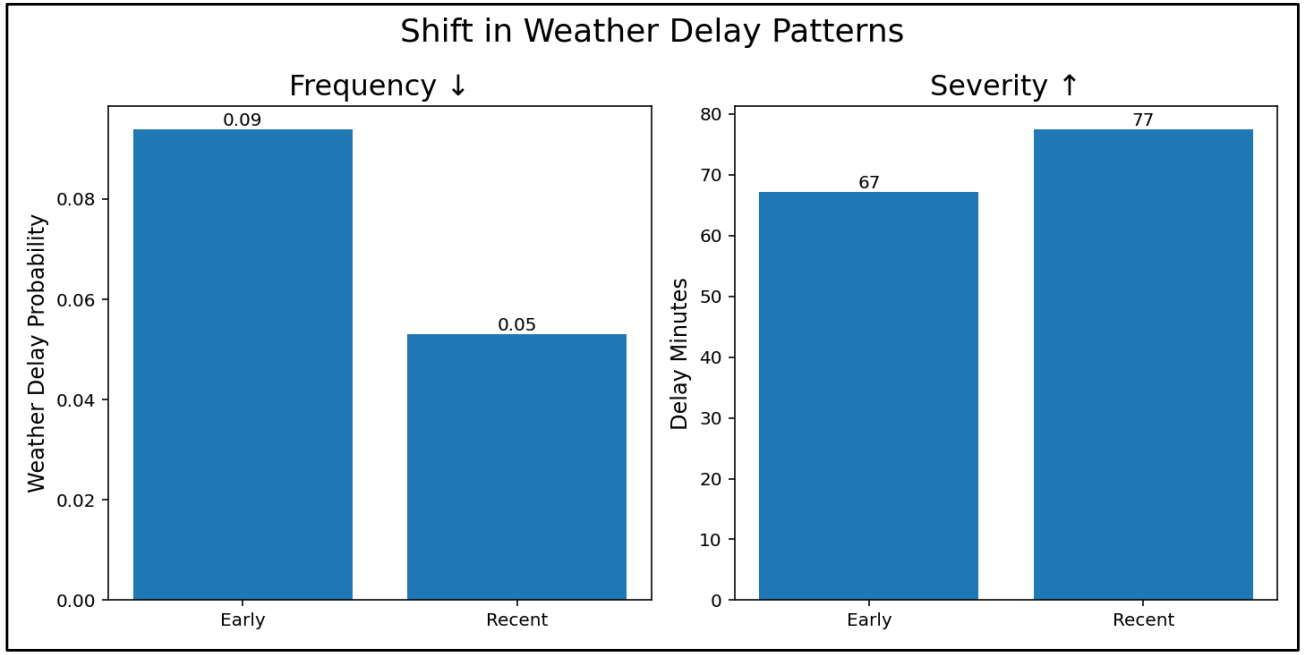




Key Takeaway

Although overall delay probability has fallen, the weather delays that do occur are getting worse — severity of the 75th percentile has climbed from ~60 to 90+ minutes and the share of months per year that have extreme weather delays has significantly increased overtime.

TECHNOLOGICAL ADVANCEMENT VS CLIMATE CHANGE



↓ Decreasing Frequency

Technological advances in forecasting, routing, and air traffic management

↑ Increasing Severity

More extreme weather events driven by climate change

The Aviation Question

Can technological progress outpace climate change?

PREDICTIVE MODELING

Example Predictions

Input: Output:

Airport	Carrier	Month	Delay Probability	Delay Minutes If Delayed	Most Likely Cause
JFK	F9	1	23%	71	Late Aircraft Delay
MIA	UA	7	30.6%	77	Late Aircraft Delay
PIT	AA	9	21.5%	63	Late Aircraft Delay
ATL	DL	12	24.8%	58	Carrier Delay